ADVANCED FILE COMMANDS

DISK USAGE AND DISK FREE

du <options>

- Allows a user to his/her disk usage including usage in subdirectories
- By default, the disk usage is presented in 1024 bytes per block

df <options>

- Shows the disk statistics of a file system
- Can show used and available space, and percentage usage

ADVANCED FILE COMMANDS

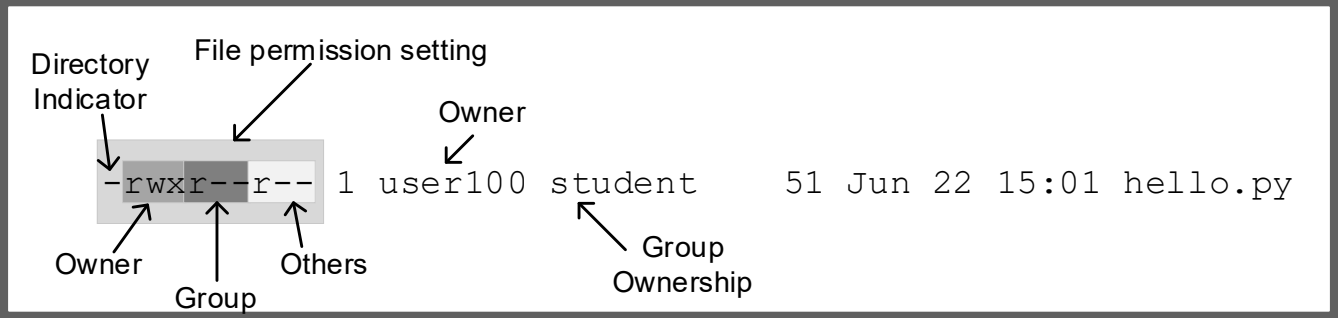
ln [target file] [link name]

- Used to create alternative names linked to a file or directory
  - Hard link (default)
    - Changes metadata only to add a new file name for the same file
    - Not able to link across file systems
    - File is accessible as long as at least 1 hard link exists
  - Soft / Symbolic link (-s option)
    - creates a new file that contains a link to the target file or directory
    - Able to link across file systems
    - Easy to identify when listing using ls command
    - No longer usable if original file is deleted

FILE PERMISSION

- Each file in the Unix system has an owner and permission
  - Each file is associated to an owner using its user ID
- A group may also have ownership of a file
  - A group is a logical grouping of users
- The owner has access to the file and modifies the permission of a file
- File permission is way to provide file security in the Unix system

FILE PERMISSION



- File has permission setting for owner, group or others (also known as world)
  - These settings can be different from each other
- The rwx stands for read, write and execute, these are toggle bits
  - Is also seen as an octal number
- File permissions are also applicable to directories

SETTING PERMISSIONS

chmod vyz <file name>

- The command uses integers to change the permission of a file
  - Where vyz is the permission integer (in octal) to be put into the file
- The v, y, and z represents the permission for owner, group and world respectively
- The permission uses toggle bits for read, write and execute

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